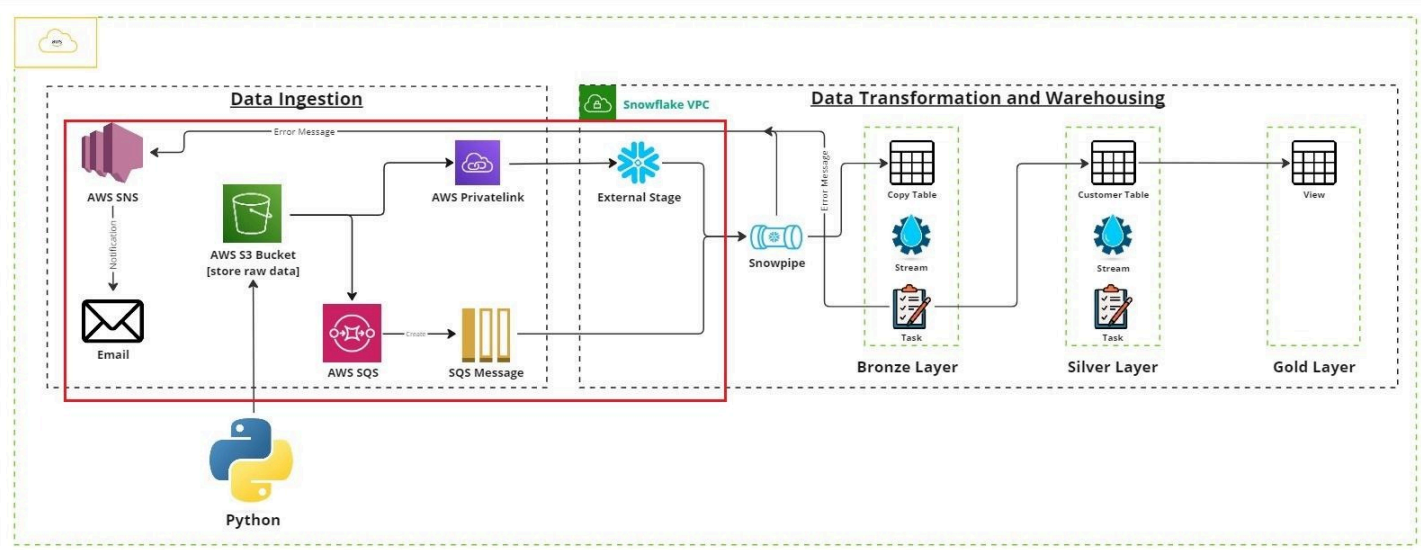
Technical Overview

In this document, we are going to set up S3 Bucket in AWS Console. Along with setup, integration with Snowflake will also be done.

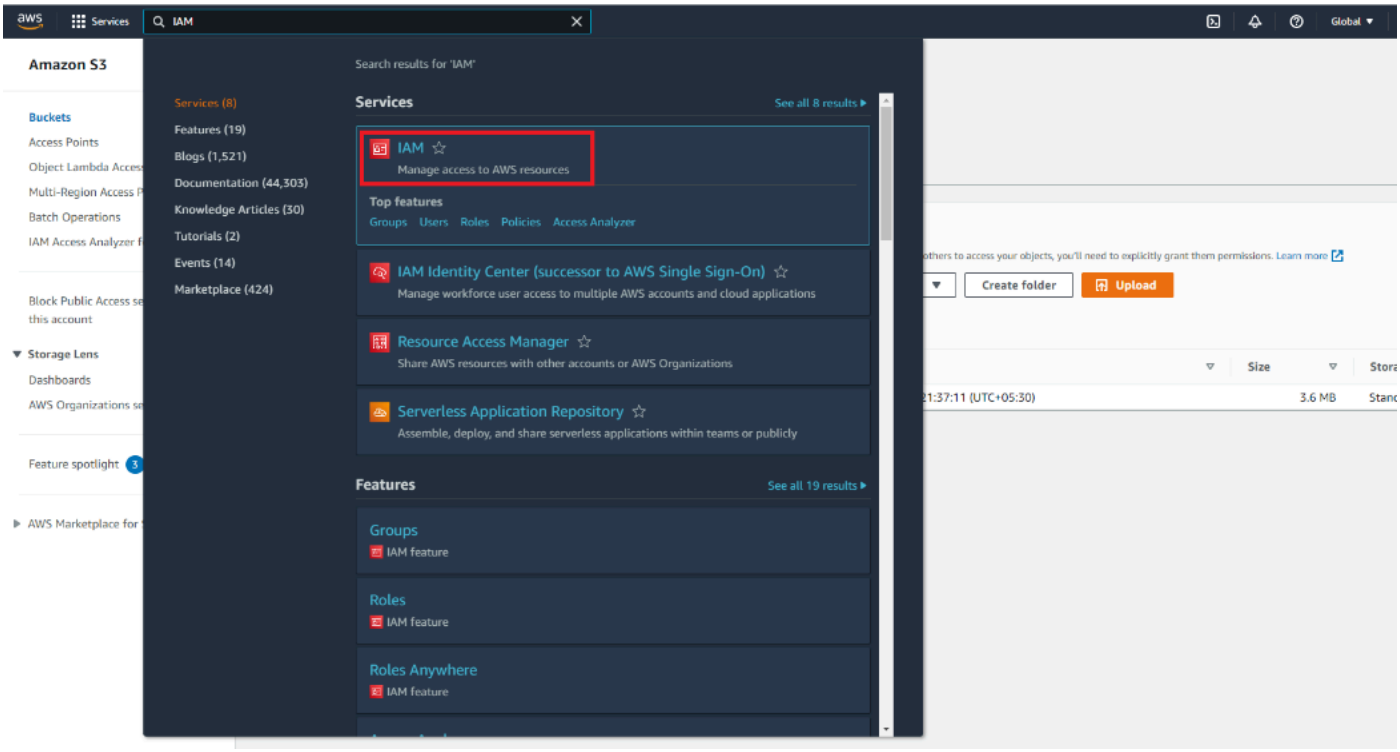


Pre-requisites:

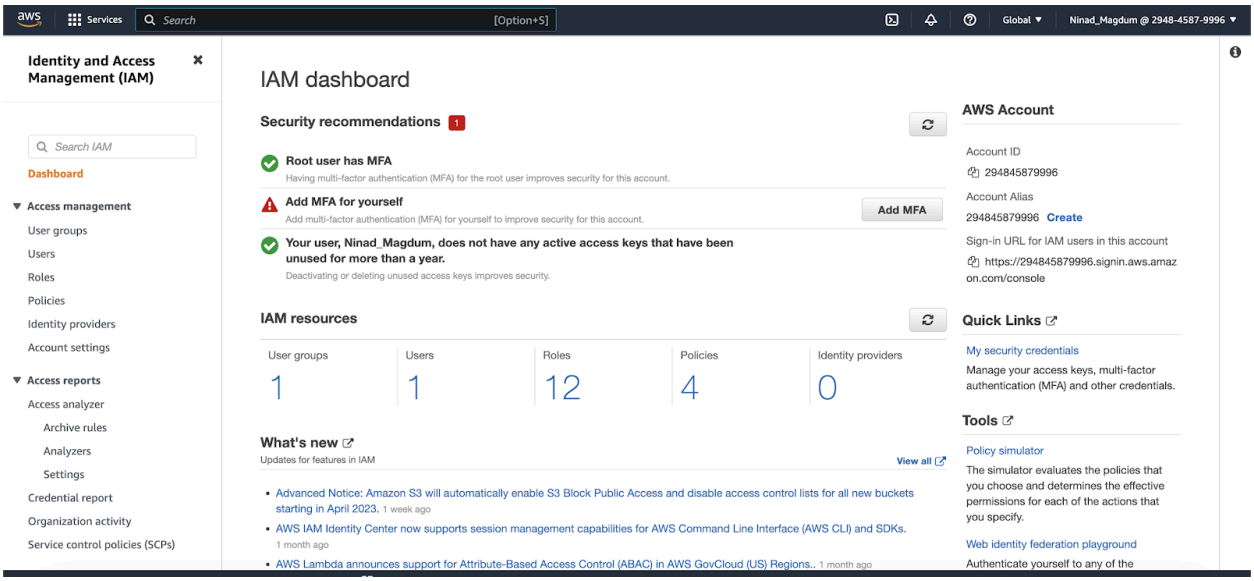
- AWS Account
- Snowflake Account

Snowflake Role Creation

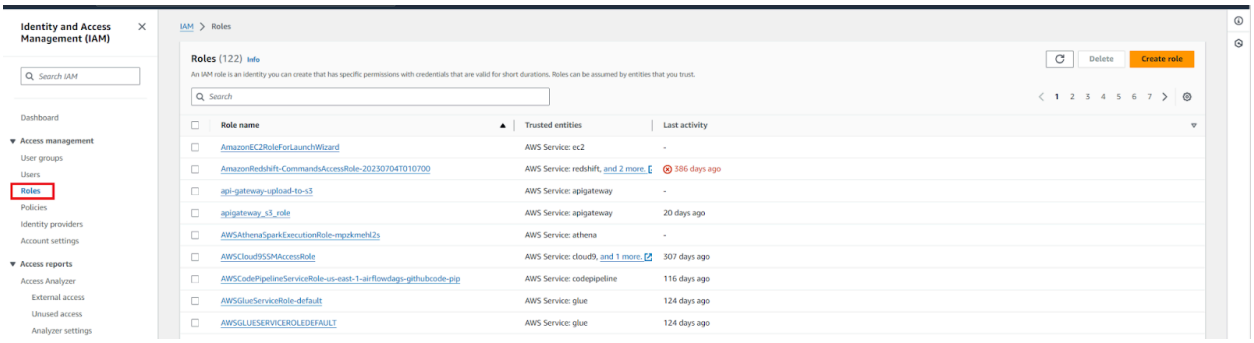
- Open your AWS Management Console and navigate to AWS IAM (you can search for "IAM" in the search bar).



- Navigate to IAM, where you'll land on the page below. Here, you'll create the necessary snowflake role



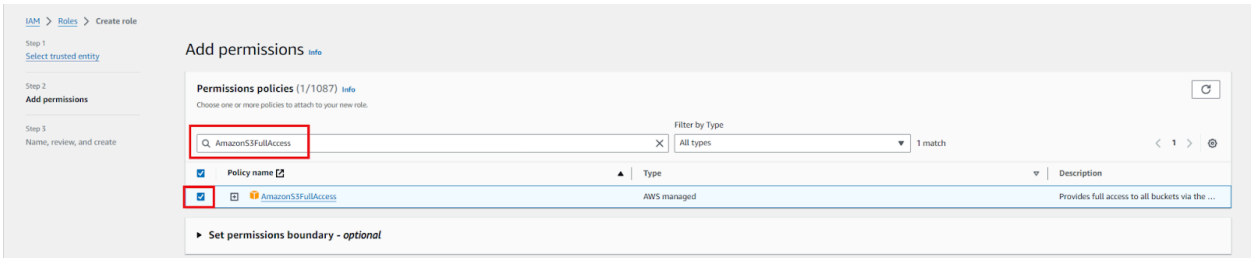
- Navigate to the “Roles” button located on the left-hand side of the page, which will direct us to the following page shown below.



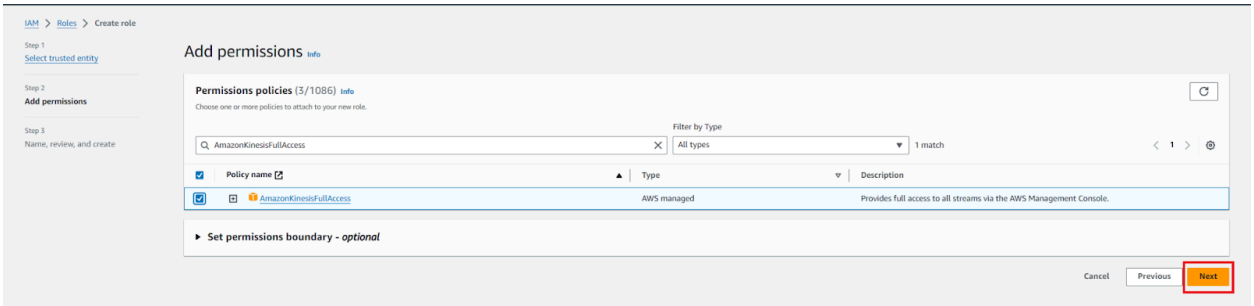
- Next, click on the "Create role" button.

- On the next page, choose the AWS account option and check the Require external ID and pass the dummy value of 0000 in the External ID box as indicated below and click Next.

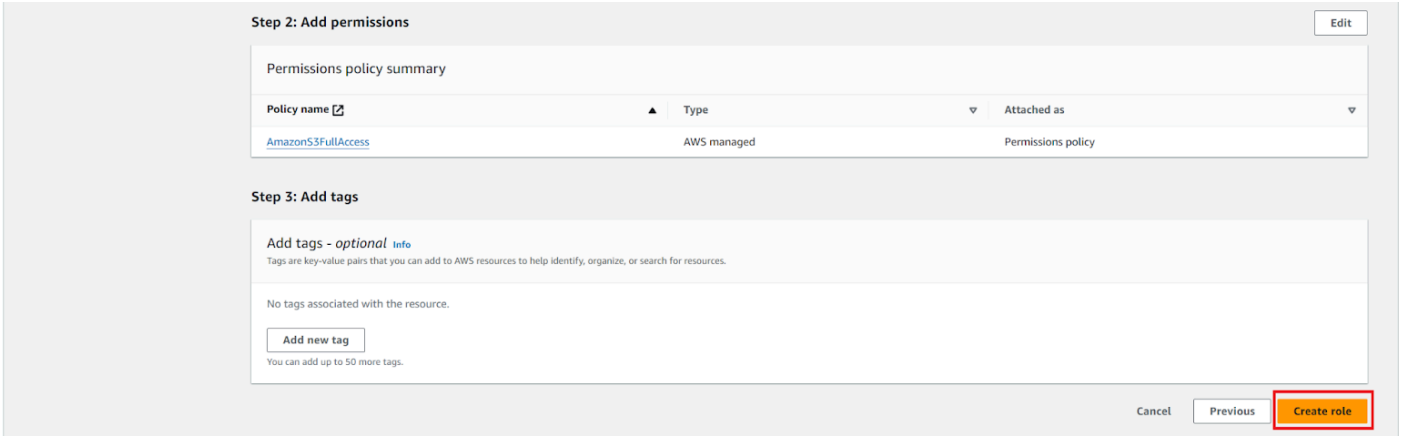
- On the following page for ADD permissions, please locate the below policy and select to include them in your role.
  - AmazonS3FullAccess



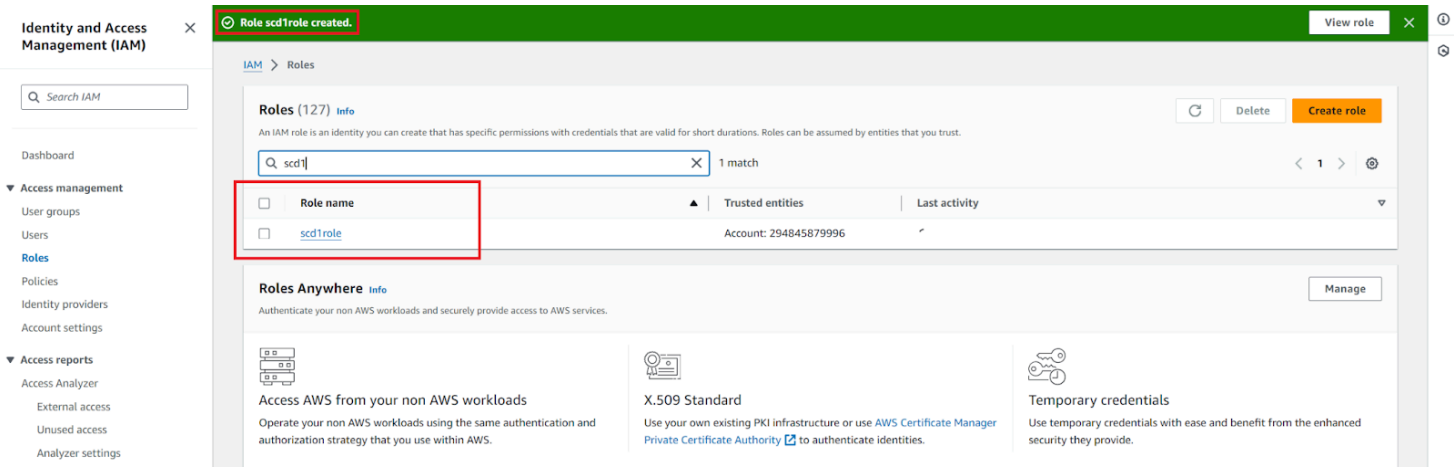
- After selecting the policies, click on the "Next" button.



- On the following page, provide an appropriate name for the role <scd1role>, then click the "Create role" button.



- The required role is created successfully

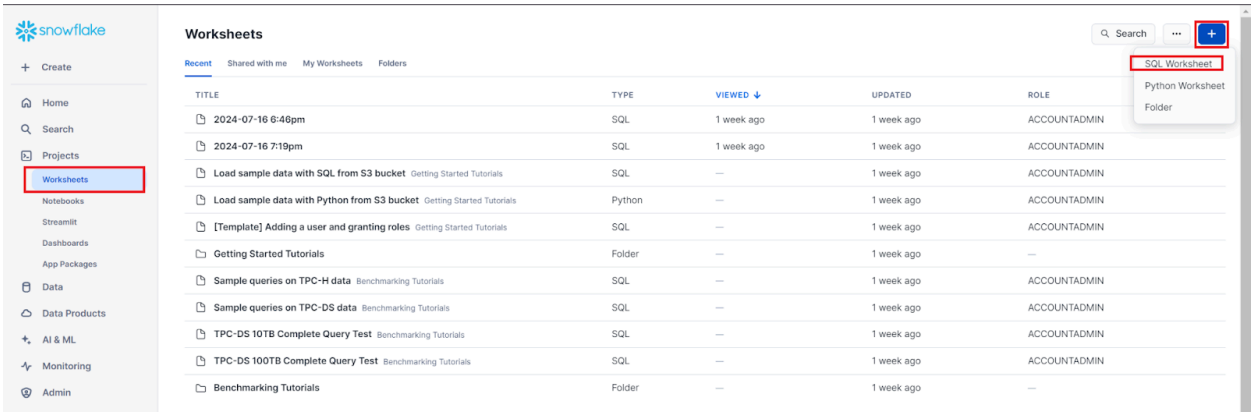


Cloud Storage Integration in Snowflake

- Go to the snowflake console

- Now Click the Projects button present on the left hand side.

- Click the Worksheets option under the Projects. Click the + button next and then click SQL Worksheet to create a new sql worksheet.



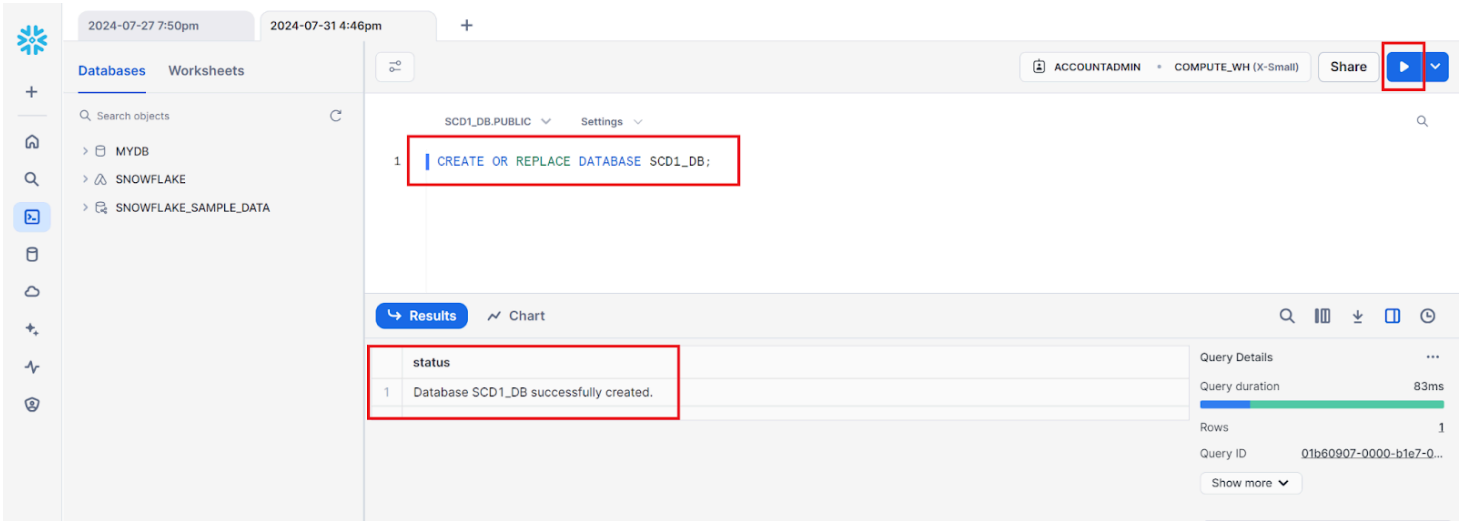
- Once you are in the SQL worksheet page execute the below two SQL statements to select your role and warehouse accordingly by clicking the play button on the top right corner

USE ROLE ACCOUNTADMIN;

USE WAREHOUSE COMPUTE_WH;

- Now setup a database to store the data by executing the below query

```
CREATE OR REPLACE DATABASE SCD1_DB;
```

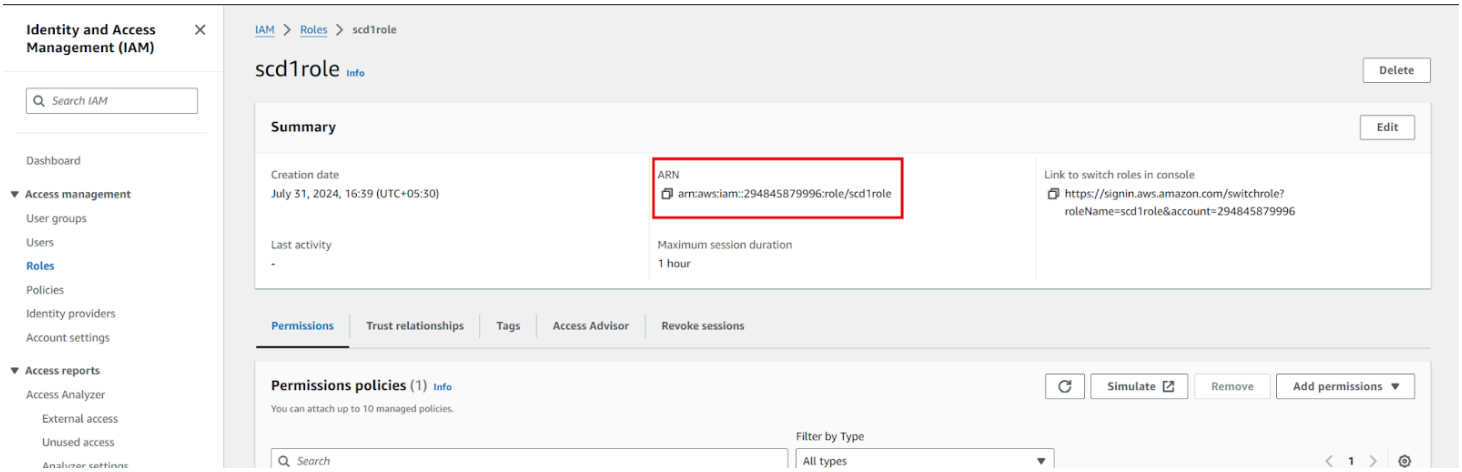


- Once the above sql statement is executed successfully,please refresh the page and validate that you have the newly created database under the database list on the left hand side with a PUBLIC schema under it.

- Now create the storage integration by running the below SQL command in your snowflake worksheet after replacing the values of STORAGE_AWS_ROLE_ARN and STORAGE_ALLOWED_LOCATIONS below with your details.

```
CREATE OR REPLACE STORAGE INTEGRATION SCD1_INT
  TYPE = EXTERNAL_STAGE
  STORAGE_PROVIDER = 'S3'
  ENABLED = TRUE
  STORAGE_AWS_ROLE_ARN = 'Your_scd1_role_arn_from_aws'
  STORAGE_ALLOWED_LOCATIONS = ('Your_S3_URI_of_data_bucket_with_folder');
```

- Add the scd1 role arn that you have created in the earlier step in the above query form the below location in your AWS



- Add the S3 data bucket folder URI in the above query form the below location in your AWS

- After clicking the play button, your S3 storage integration will be created successfully.

- Run the below sql command to describe the SCD1 Storage integration details by clicking the play button and capture the below property values

- STORAGE_AWS_IAM_USER_ARN
- STORAGE_AWS_EXTERNAL_ID

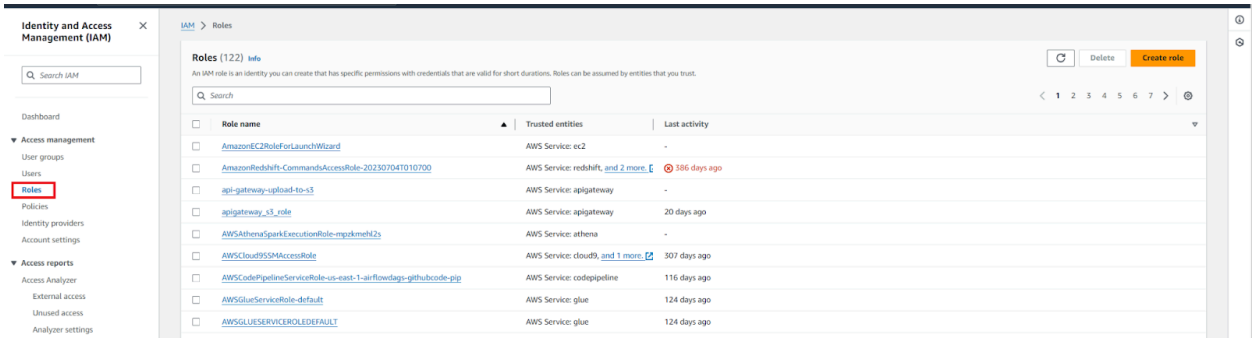
DESC INTEGRATION SCD1_INT;

- Now let’s update the Trust relationships of the scd1 role in AWS IAM

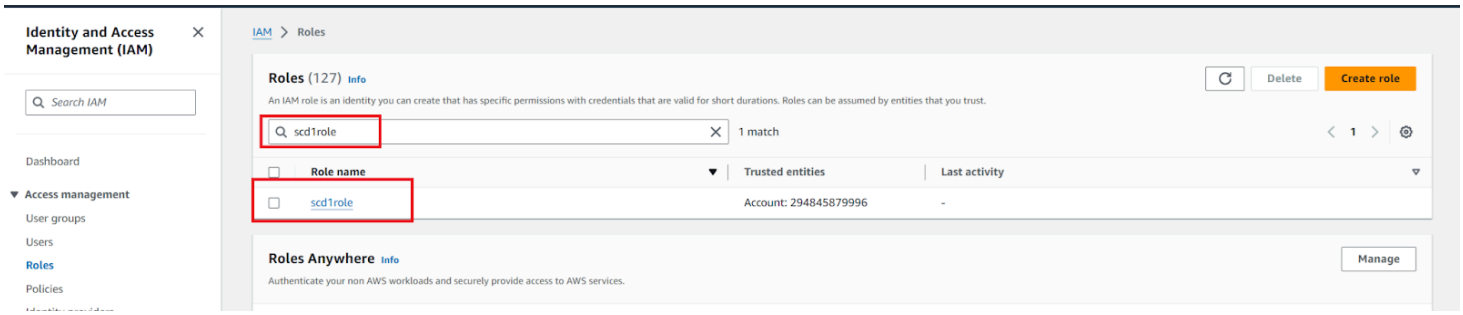
- Open your AWS Management Console and navigate to AWS IAM (you can search for "IAM" in the search bar).

- Navigate to IAM, where you'll land on the page below.

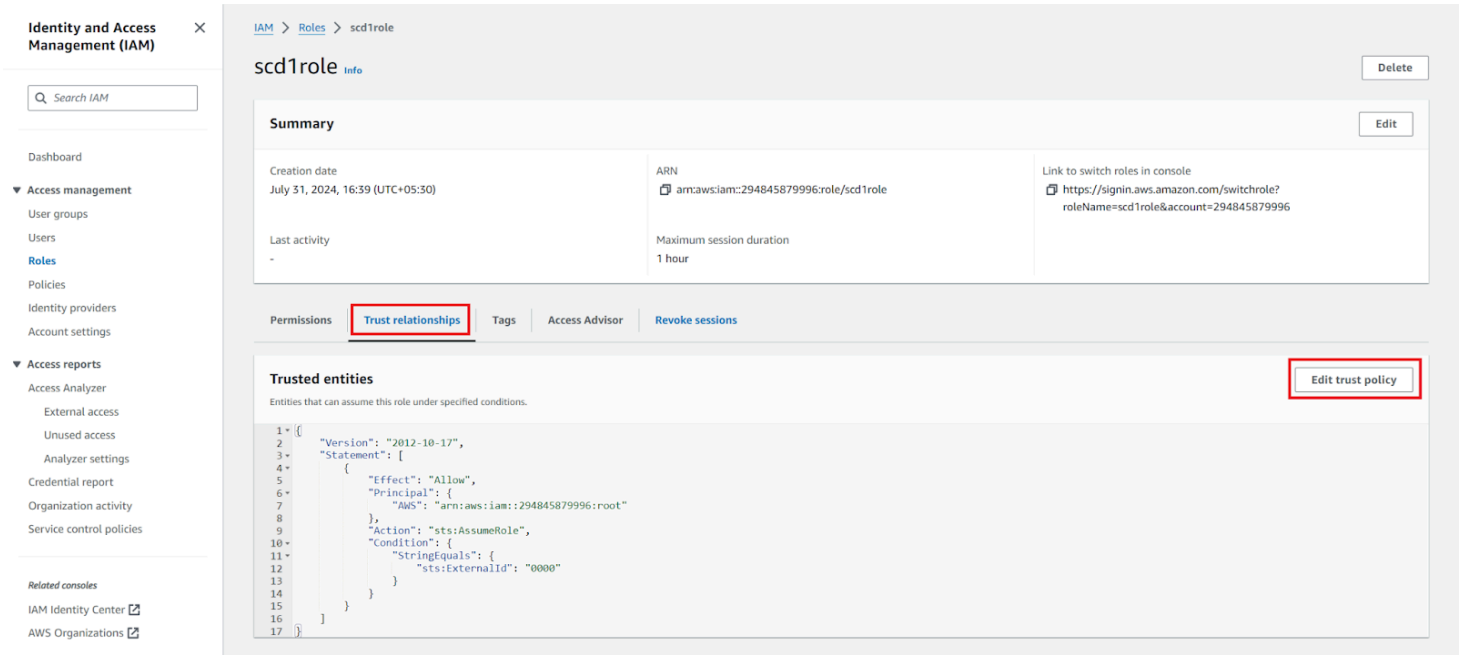
- Navigate to the “Roles” button located on the left-hand side of the page, which will direct us to the following page shown below.



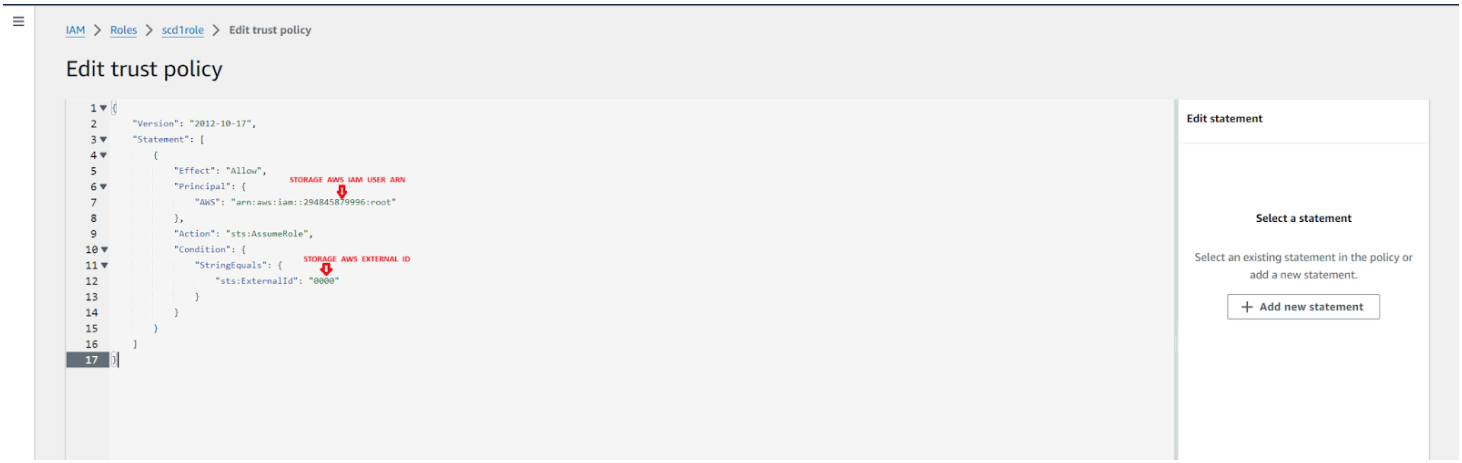
- Search the newly created scd1 role in the search bar and click it to open it.



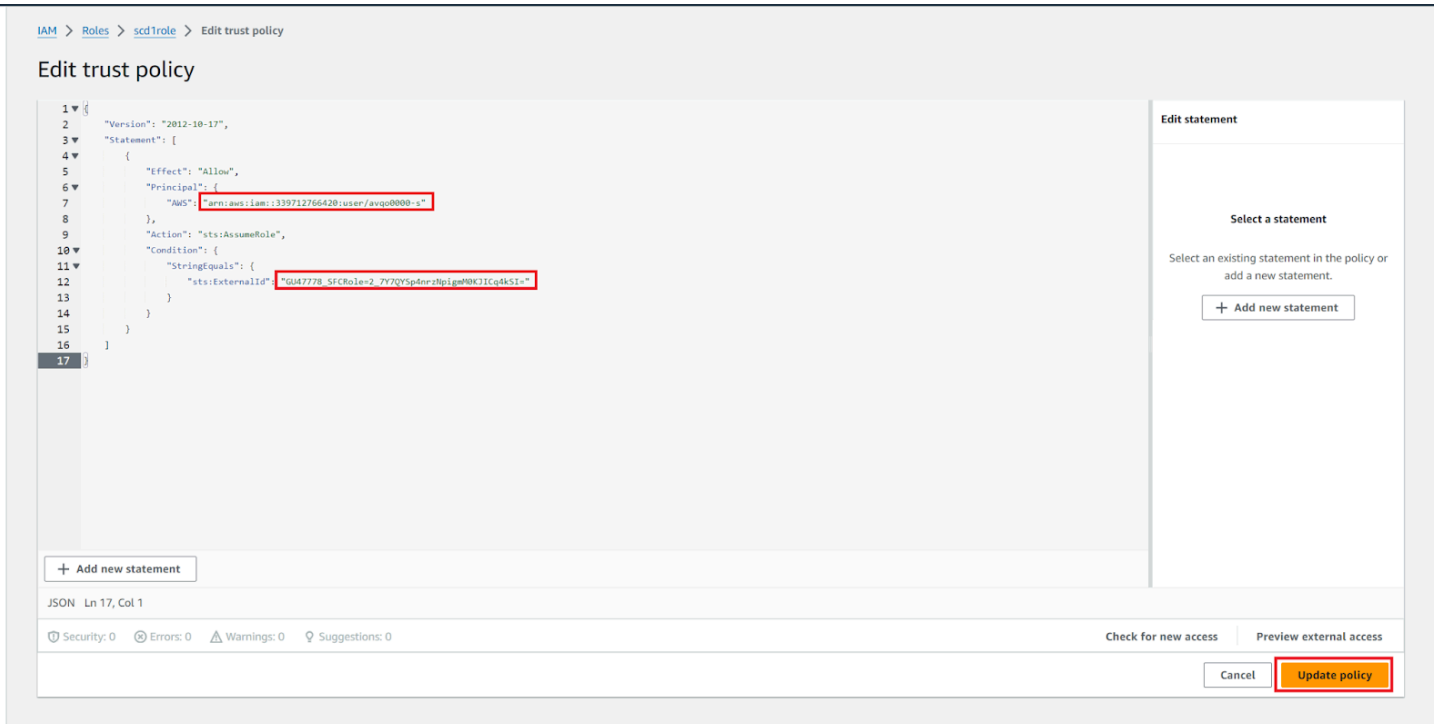
- Click on the Trust relationship tab in the next page and click the Edit trust policy button to update the existing Trusted entities condition



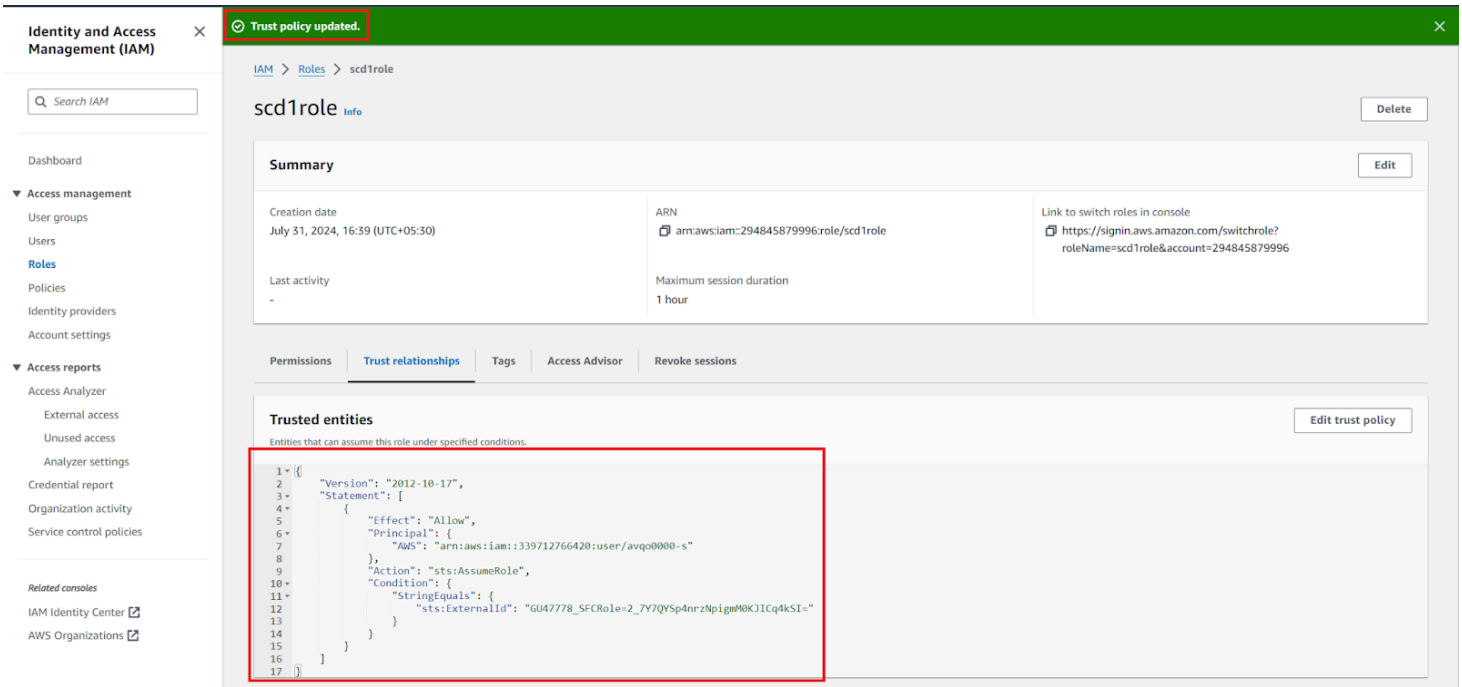
- Replace Existing AWS value with STORAGE_AWS_IAM_USER_ARN and ExternalId with STORAGE_AWS_EXTERNAL_ID values fetched from snowflake in previous step in the Trusted entities



- Once you update the values, click the Update policy button to update the Trust policy.



- Your role Trust policy is updated now.



Create a stage in Snowflake

- Execute the below code in the snowflake worksheet to create a stage after replacing the value of URL below with your details.

```
CREATE OR REPLACE STAGE SCD1_DB.PUBLIC.SCD1_STAGE

STORAGE_INTEGRATION = SCD1_INT

URL= 'Your_S3_URI_of_data_bucket_with_folder';
```

- Add the S3 data bucket folder URI in the above query form the below location in your AWS

- After clicking the play button, your SCD1 Stage will be created successfully.

- Execute the below SQL query to verify if the stage integration is set up correctly. If no errors occur after running the query, the setup is successful. If errors do appear, please revisit the previous setup steps.

```
ls @SCD1_DB.PUBLIC.SCD1_STAGE;
```

Successfully setup of AWS S3 Bucket and the integration of the same with Snowflake

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